Digital Forensic Investigation Report

CIS607-T301 Computer Forensics.

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Final Case Study.

Case Information

Case Name: Jean Case – M57.biz
Case Number: DF-2025-08
Investigator: Digital Forensics Analyst Ezra Shuremu
Date: 2025-08-05
Tool Used: Autopsy 4.17.0

Objective

The purpose of this forensic investigation is to analyze the digital evidence from the M57.biz case, identify any unauthorized email communications, detect potential spoofed emails, and document the chain of events that led to the unauthorized disclosure of sensitive employee information.

Evidence Overview

Evidence Item	Description	Source
outlook.pst	Microsoft Outlook PST file containing email communications	img_nps-2008-jean.E01
Keyword Searches	Email addresses, spoofed emails, and sensitive data-related keywords	Autopsy Keyword Search

Analysis

During the investigation, multiple email accounts associated with the suspect machine were identified. Among these, a spoofed email was discovered that was inserted into the communication chain. This email, appearing to be from Alison (alison@m57.biz), was

actually sent from tuckgorge@gmail.com. The spoofed email requested sensitive employee information including names, salaries, and Social Security Numbers (SSNs). The following screenshot shows the spoofed email details identified within Autopsy.

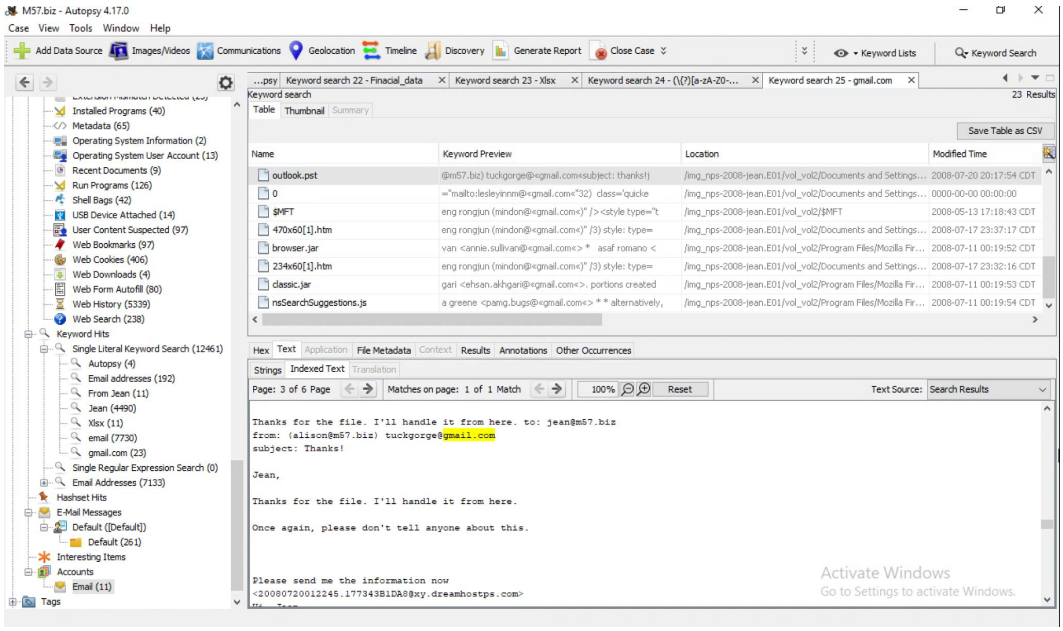

Figure 1: Spoofed email identified in Autopsy.

Findings

Key findings from the investigation include:

- Multiple email accounts were discovered from the Outlook PST file.
- A spoofed email was identified, requesting sensitive data.
- Keyword searches confirmed financial and email-related queries.
- Metadata of the email artifacts matched the timeline of suspicious activity.

Timeline of Events

Date/Time (CDT)	Event	Details
2008-07-20 20:17:54	Spoofed Email Sent	Spoofed email from tuckgorge@gmail.com to jean@m57.biz requesting sensitive employee data
2008-07-20 20:17:54	PST File Modified	Email communications stored in outlook.pst updated

Conclusion

The forensic analysis revealed that Jean was targeted with a spoofed email that led to the unauthorized disclosure of sensitive information. The email originated from an external Gmail account and was designed to appear as if it was sent from a trusted internal contact. Evidence was collected using Autopsy and documented properly for potential legal proceedings.

Recommendations

- Implement email spoofing detection mechanisms (SPF, DKIM, DMARC).
- Provide user awareness training on identifying phishing and spoofed emails.
- Regularly audit email server logs and conduct forensic readiness exercises.
- Report the incident to relevant legal and compliance departments.